

EDUGENIE

5.1 Experiential & Constructivist Learning Models

Constructivist learning emphasizes that learners actively construct knowledge through experience. EDUGENIE supports this approach by allowing students to interact with environments instead of only receiving information.

For example, instead of explaining bridge design only through a textbook, students could construct a virtual bridge, test it, identify weaknesses, and modify their design.

The learning process can follow: Explore → Experiment → Observe → Reflect → Improve. Teachers can also create problems that require students to form hypotheses and test them.

Key Points

- Encourages active participation.
- Supports problem-solving.
- Promotes curiosity.
- Connects theory with practical experience.

Conceptual Picture / Workflow

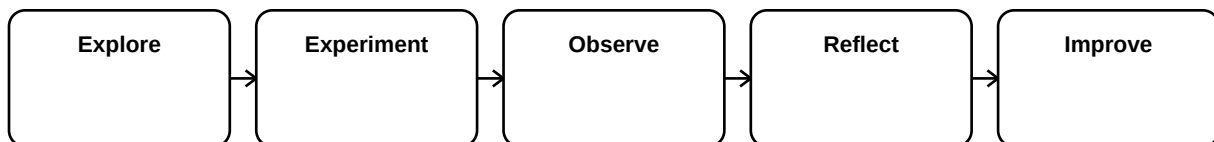


Figure: Simplified representation of the learning mechanism.

Comparison / Design Table

Element	Traditional approach	EDUGENIE approach
Learning	Receive information	Explore and construct knowledge
Practice	Fixed examples	Interactive experimentation
Feedback	Often delayed	Can be immediate
Reflection	After activity	Built into the learning cycle

Sample Coding Concept

```
# Conceptual learning loop
world = create_virtual_bridge()

while student_is_learning:
    action = student.choose_action()
```

```
result = world.simulate(action)
student.observe(result)
student.reflect()

if student.needs_revision():
    world.reset_and_try_again()
```

Example Classroom Activity

A teacher can ask students to design a bridge that supports a target load. Students form a hypothesis, build the virtual structure, test it, observe failure points, modify the design, and explain why the improved design performs differently.